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Exponential derivatives and generalized reproduction factors

A short introduction

Abstract. Given a differentiable path $c : \mathcal{I} \to G$ in a Lie group, the exponential derivative of c acts through right or left multiplication *within* G and offers so an *intrinsic* description of the infinitesimal change, in contrast to the classical derivative $\dot{c} = \mathrm{dc/dt}$ that acts in *tangent space* $TG_{c(t)}$. An elementary example is provided by the multiplicative group $G = \mathbb{R}_+$ together with the smooth strictly positive functions $f : \mathcal{I} \to \mathbb{R}_+$. This is treated in more detail as a prototype.

Basic properties, similarities and differences to the classical derivative are outlined and calculation rules are listed. In the Appendix, an application to Mathematical Epidemiology is presented.

Keywords: Lie groups, exponential derivative, reproduction factor

MSC-classes: 26A06, 37C05, 92D30

1. Introduction

The exponential derivative is not new. In fact, it is extensively applied in these days of the Corona pandemic without being explicitely mentioned as such. Instead, we encounter the term *reproduction factor* almost every day in the news, and it tells us how many people are infected on average from one contagious person, cf. [vDr]. One might wonder what this notion has to do with a *derivative*. There is no explicit time scale visible in its definition and so a connection to a time derivative does not seem to make sense. But time enters naturally with the *transmission time*, which is the average time T_{tr} during that an epidemic with reproduction factor R spreads from one infected person to R susceptible ones. The fraction $u(t)$ of newly infected individuals per time unit, the so-called incidence, then increases or decreases with time according to

$$u(t + \Delta t) \approxeq R_t^{\Delta t/T_{tr}} \cdot u(t) \tag{1.1}$$

in higher than first order with $\Delta t \to 0$, thus introducing a natural time scale for the epidemic spread. The mapping $R_t^{\frac{x}{T_{tr}}} = \exp\left[\, x \, \frac{du}{dt}/u(t) \,\right]$ is in this case the exponential derivative of the incidence $u(t)$. As an *analytic one-parameter subgroup*, it approaches u at time t within the *multiplicative* group $\mathbb{R}_+$, rather than in the *additive* group of the tangent space (isomorphic to $\mathbb{R}$), like the classical differential.

Relation (1.1) calls for generalization and that is what this study is essentially about. As will be seen, Lie groups provide the natural framework. However, in an introductory study, a reliable bridge should be built to what is well known, and therefore we treat the elementary case $G = \mathbb{R}_+$ in some detail.

The exponential derivative expresses the infinitesimal change of a C^∞-path $c : \mathcal{I} \to G$ by inner multiplication, i.e. using a factor *in* G , in contrast to the classical differential which acts by addition in *tangent space*. Consistent with that, equation (1.1) can be read as a *'finite quotient'* instead of a finite difference equation.

In an elementary abelian model, for instance based on $G = \mathbb{R}_+^n$, the exponential derivative may describe a chemical reaction, the energy freed in a chain reaction, or the evolution of a pandemic, for example. Typically, the dynamic equations exhibit transient exponental increase or decrease with a time-dependent base (as exemplified in the Appendix).

Various applications are conceivable for the general Lie group framework. We are particularly thinking of dynamical systems with many interacting components.

Last but not least, the whole theory is so *natural, clear and simple* (not to say self-evident in many steps, especially in its elementary setting, cf. Section 4), that one might wonder why it is not part of any advanced Calculus course.

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2. Basic ideas

The terms *differentiable*, *smooth*, or C^∞ will be used synonymously with *differentiable in every order*. For instance, a differentiable path is a C^∞-path and a differentiable manifold is a C^∞-manifold as defined in standard textbooks of Differential Geometry, e.g. [Ddn], [Jst], [Hel].

In classical Analysis, the differential of a C^∞-path $c : \mathcal{I} \to \mathbb{R}^n$ in $t \in \mathcal{I}$ is a linear map $Dc_t(x) : \mathbb{R} \to \mathbb{R}^n$ and in particular a homomorphism from the additive group of $\mathbb{R}$ into that of $\mathbb{R}^n$ such that

$$\frac{d}{ds}\, c(s)\big|_t \;=\; \frac{d}{dx}[\, Dc_t(x) + c(t)\,]\big|_0 \tag{2.1}$$

which implies that $Dc_t(x) + c(t)$ approximates c in t in higher than the first order, that is

$$c(t + \Delta t) \;=\; Dc_t(\Delta t) + c(t) \;+\; O(\Delta t) \quad . \tag{2.2}$$

In analogy to (1.1) this is written

$$c(t + \Delta t) \;\approxeq\; Dc_t(\Delta t) + c(t) \quad . \tag{2.3}$$

In these equations, the differential Dc_t acts *directly in* $\mathbb{R}^n$ thanks to the canonical identification of $\mathbb{R}^n$ with its tangent space. The situation is obviously different in a differentiable manifold where no such natural identification exists, quite apart from the fact that there is usually no addition or another binary operation in an arbitrary C^∞-manifold, with which equations (2.2) and (2.3) would make sense. Again, things are different, if the manifold is a Lie group G. Then the differential Dc_t is still operating linearly in the tangent space $TG_{c(t)}$, however, via the exponential map $\exp : \mathfrak{g} \to G$, the differential Dc_t gives rise to an *analytic one-parameter subgroup of* G, i.e. an analytic homomorphism $Ec_t(x) : \mathbb{R} \to G$ into the connected component of the unit element $\mathbf{1}_G \in G$, such that $Ec_t(x)$ acts from the left (or alternatively from the right) *directly in* G and $Ec_t(\Delta t) \cdot c(t)$ approximates $c(t+\Delta t)$ for $\Delta t \to 0$. The latter (since we are in general not dealing with normed space) in the precise sense that

$$\frac{d}{ds}\, c(s)\,\big|_t \;=\; \frac{d}{dx}[\, Ec_t(x) \cdot c(t)\,]\big|_0 \tag{2.4}$$

in correspondence to (2.1). In that case we write in analogy to (2.3)

$$c(t + \Delta t) \;\approxeq\; Ec_t(\Delta t) \cdot c(t)\,. \tag{2.5}$$

Based on this formal correspondence, which is made precise in the following section, $Ec_t(x)$ is named the (*left*) *exponential derivative of* c *at* t.

Notice that relation (2.4) extends naturally to C^∞-paths $c : \mathcal{I} \to M$ in a differentiable *manifold* if the connected component of $\mathbf{1}_G$ of the Lie group operates diffeomorphically on M (or at least on an open environment of $c(\mathcal{I})$).

From now on we just write simply gh instead of $g \cdot h$ for the binary operation in G.

3. Geometric analysis

Given a Lie group, i.e. a differentiable manifold G with an analytic binary operation which makes G a group and given a differentiable path $c : \mathcal{I} \to G$ defined on an interval $\mathcal{I} \subset \mathbb{R}$, we first state the following

THEOREM 1

1. For every $t \in \mathcal{I}$, there exists an unique analytic homomorphism $Ec_t : (\mathbb{R}, +) \to G$, $x \mapsto Ec_t(x)$, i.e. an *analytic one-parameter subgroup of* G, such that

$$\frac{d}{ds}\, c(s)\big|_t \;=\; \frac{d}{dx}\,[\, Ec_t(x)\, c(t)\,]\big|_0\,. \tag{T1-1}$$

2. The mapping

$$\begin{aligned} Ec : \mathcal{I} \times \mathbb{R} &\longrightarrow G \\ (t, x) &\longmapsto Ec_t(x) \end{aligned}$$

is differentiable. (T1-2)

In particular, for every fixed $x \in \mathbb{R}$ is the mapping $Ec : t \mapsto Ec_t(x)$ a differentiable path $\mathcal{I} \to G$.

Proof : Define the function

$$\begin{aligned} \mathbf{c} \,:\, \mathcal{I}\times\mathcal{I} &\longrightarrow G \\ (s,t) &\longmapsto c(s)\,c^{-1}(t)\ . \end{aligned} \tag{3.1}$$

Then $\mathbf{c}$ is obviously differentiable as a composition of differentiable maps. Also, $\mathbf{c}(t,t) = \mathbf{1}_G$ (the unit element of G), hence $\frac{d}{ds}\mathbf{c}(s,t)\big|_t = \frac{d}{ds}\,c(s)\,c^{-1}(t)\big|_t \in TG_{\mathbf{1}_G} = \mathfrak{g}$ and the exponential map

$$\begin{aligned} Ec_t(x) \,:\, \mathbb{R} &\longrightarrow G \\ x &\longmapsto \exp\big(\,x\frac{d}{ds}\mathbf{c}(s,t)\big|_t\,\big) \end{aligned} \tag{3.2}$$

is well defined for every $t\in\mathcal{I}$ and is an analytic homomorphism of the additive group of $\mathbb{R}$ into G. According to (3.2), $Ec_t(x)$ satisfies

$$\frac{d}{ds}c(s)\,c^{-1}(t)\big|_t \;=\; \frac{d}{dx}Ec_t(x)\big|_0\,. \tag{3.3}$$

The latter is however equivalent to equation (T1-1), since for every $g\in G$ the right action of g on G is an analytic diffeomorphism, hence in particular the action of $g = c(t)$ for every fixed t. The uniqueness of $Ec_t(x)$ then follows from the uniqueness of the exponential map $\exp(x\,\mathbf{v})$, which as an analytic homomorphism of $\mathbb{R}$ into G satisfies $\frac{d}{dx}\exp(x\,\mathbf{v})\big|_0 = \mathbf{v}$, for every $\mathbf{v}\in\mathfrak{g}$ (cf. [Hel], chap II, Corollary 1.5).

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If $\mathcal{C}$ denotes the Lie group of all differentiable paths $c:\mathcal{I}\to G$ and $\mathrm{Hom}\,(\mathbb{R},G)$ the set of analytic homomorphisms from $\mathbb{R}$ into G, the relationship between Ec_t and the exponential map is graphically summarized in this commutative diagram

$$\begin{tikzcd} & \mathrm{Hom}\,(\mathbb{R},G) \\ \mathcal{C}\times\mathcal{I} \arrow[ur, "Ec_t"] \arrow[r] \arrow[d, leftrightarrow, "(c,t)\mapsto \frac{d}{ds}c(s)\,c(t)^{-1}\big|_t"'] & \mathfrak{g} = TM_{\mathbf{1}_G} \arrow[u, "\exp"'] \\ \end{tikzcd}$$

Definition 1

The unique function $Ec : t\mapsto Ec_t(x) \in \mathrm{Hom}\,(\mathbb{R}\,,G)$ satisfying equation (T1-1) of Theorem 1 is called the (*left*) *exponential derivative* of c in t.

Since we deal only with the left exponential derivative in the following, the adjective *left* can be omitted from now on. The 'right case' is treated analogously.

As usually, let the *right* and *left translation* and the *conjugation* with $g\in G$ be defined by

$$\begin{aligned} R_g : G &\longrightarrow G\,, \\ h &\longmapsto hg \end{aligned} \qquad \begin{aligned} L_g : G &\longrightarrow G \\ h &\longmapsto gh \end{aligned} \qquad \text{and} \qquad \begin{aligned} \phi^g : G &\longrightarrow G\,, \\ h &\longmapsto \phi^g(h) := g\,h\,g^{-1} \end{aligned} \qquad \text{respectively}\,.$$

As is well known, R_g and L_g are analytic *diffeomorphisms* and $\phi^g(h)$, also simply written h^g, is an analytic *endomorphism* of G. With these objects, the following derivation rules hold

THEOREM 2 (BASIC RULES) .

Let $g\in G$ and $c, c^{\sim} : \mathcal{I}\to G$ be differentiable paths.
The following is easily veryfied:

1. If $c = const$, then $Ec = \mathbf{1}_G$ (i.e. the trivial homomorphism $\mathbb{R}\ni x\mapsto \mathbf{1}_G$) (T2-1)
2. For every $n\in\mathbb{N}$, $Ec^n = (Ec)^n$. (T2-2)
3. If c, $c^{\sim}$ commute, then $Ec\,c^{\sim} = Ec\,Ec^{\sim}$. (T2-3)
4. $ER_g\,c = Ec$ (T2-4)
5. $EL_g\,c = Ec^g \qquad (\,=\, E\phi^g(c)\,)$. (T2-5)

It is obvious that, if $c :]-\infty, \infty[\to G$ is a differentiable one-parameter group, i.e. $c(r+s) = c(r)\,c(s)$ for every $r, s \in \mathbb{R}$, then $Ec_t \equiv c$ for every $t \in \mathbb{R}$.

Corollary to (T2-3): If G is abelian, then the mapping $c(t) \mapsto Ec_t(1)$ is an endomorphism of the group of differentiable paths $c : \mathcal{I} \to G$.

Proof: We prove statement (T2-3), the remainder is left to the reader.
Let $t \in \mathcal{I}$. Then, since c, $c^{\sim}$ commute,

$$\begin{aligned} Ecc^{\sim}_t &= \exp\{\, x \frac{d}{ds}\,[\,c(s)\,c^{\sim}(s)\,c^{\sim}(t)^{-1}\,c(t)^{-1}\,]\big|_t\,\} \\ &= \exp\{\, x \frac{d}{ds}\,[\,c(s)\,c(t)^{-1}\;c^{\sim}(s)\,c^{\sim}(t)^{-1}\,]\big|_t\,\}\,. \end{aligned} \tag{3.4}$$

Now we apply the following

Lemma. Let $c_1, c_2 : \mathcal{I} \to G$ be differentiable paths and $t \in \mathcal{I}$, such that $c_1(t) = c_2(t) = \mathbf{1}_G$. Then

$$\frac{d}{ds}\,c_1\,c_2\big|_t = \frac{d}{ds}\,c_2\,c_1\big|_t = \frac{d}{ds}c_1\big|_t + \frac{d}{ds}c_2\big|_t \quad . \tag{3.5}$$

Note that (3.5) still applies to non-commuting paths c_1, c_2. In fact, it is easily seen that the differentiable mappings

$$\begin{array}{lll} f : G \times G \to G, & g : \mathcal{I} \to G \times G, & g_1 : \mathcal{I} \to G \times G, \quad g_2 : \mathcal{I} \to G \times G, \\ \quad (u\,,\,v) \mapsto u\,v & \quad s \mapsto (\,c_1(s)\,, c_2(s)\,) & \quad s \mapsto (\,c_1(s)\,, \mathbf{1}_G\,) \quad\; s \mapsto (\,\mathbf{1}_G\,,\, c_2(s)\,) \end{array}$$

satisfy

$$f \circ g = c_1\,c_2 \;\;,\quad f \circ g_1 = c_1 \quad \text{and} \quad f \circ g_2 = c_2\,.$$

Therefore

$$\begin{aligned} \frac{d}{ds}\,c_1\,c_2\big|_t &= df\big|_{g(t)}\,\frac{d}{ds}\,g\big|_t = df\big|_{(\mathbf{1}_G,\mathbf{1}_G)}\,\frac{d}{ds}\,g\big|_t = df\big|_{(\mathbf{1}_G,\mathbf{1}_G)}\,(\,\frac{d}{ds}c_1\big|_t\,,\,\frac{d}{ds}c_2\big|_t\,) \\ &= df\big|_{(\mathbf{1}_G,\mathbf{1}_G)}\,[\,(\,\frac{d}{ds}c_1\big|_t\,,\,0\,) + (\,0\,,\,\frac{d}{ds}c_2\big|_t\,)\,] = df\big|_{g_1(t)}\frac{d}{ds}g_1\big|_t + df\big|_{g_2(t)}\frac{d}{ds}g_2\big|_t \\ &= \frac{d}{ds}c_1\big|_t + \frac{d}{ds}c_2\big|_t\,. \end{aligned} \tag{3.6}$$

With $c_1(s) := c(s)\,c(t)^{-1}$ and $c_2(s) := c^{\sim}(s)\,c^{\sim}(t)^{-1}$ equation (3.4) becomes

$$\begin{aligned} Ecc^{\sim}_t &= \exp(\,x\,\frac{d}{ds}\,c_1\,c_2\,\big|_t\,) \\ &= \exp(\,x\,\frac{d}{ds}\,c_1\big|_t + x\,\frac{d}{ds}\,c_2\big|_t\,) \\ &= \exp(\,x\,\frac{d}{ds}\,c_1\big|_t\,)\;\exp(\,x\,\frac{d}{ds}\,c_2\big|_t\,) \quad [\,\text{since } c_1 \text{ and } c_2 \text{ commute}\,] \\ &= Ec_t\,Ec^{\sim}_t\;. \end{aligned} \tag{3.7}$$

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Remarks

Since the exponential map of the additive *Lie group* $\mathbb{R}^n$ is just the identity of $\mathbb{R}^n$, the exponential derivative of a path in $\mathbb{R}^n$, characterised by Theorem 1 (T1-1), coincedes with the classical derivative. This, besides the analogous properties listed in Theorem 2, also justifies the naming *'derivative'* in the generalized sense of Definition 1.

Apart from Differential Geometry, one finds equations of the form (T1-1) and hence some kind of "exponential derivative" also in Quantum Mechanics, where it describes the infinitesimal change of a time-dependent evolution operator $U(t)$, $t \in [\,0, \infty[$ *within* the group of unitary transformations $\mathcal{U}(\mathcal{H})$ of a separable Hilbert space $\mathcal{H}$.

In fact, for any suitably smooth family $t \mapsto H_t$ of self-adjoint energy operators with a dense common domain $\mathcal{D} \subset \mathcal{H}$ (i.e. H_t is a *time-dependent Hamiltonian*), every state vector $\phi(\,\mathbf{q}, t\,) = U(t)\,\phi(\,\mathbf{q}, 0\,) \in \mathcal{D}$ evolves in $\mathcal{H}$ according to Schrödinger's equation

$$i\,\frac{\partial}{\partial t}\,\phi\,(\mathbf{q}, t) = H_t\,\phi\,(\mathbf{q}, t)\,, \tag{3.8}$$

which yields, in analogy to (T1-1),

$$\frac{d}{ds}\,[\,U(s)\,\phi(\mathbf{q},\,0\,)\,]\big|_t = \frac{d}{dx}\,[\,\exp(\,\frac{x}{i}H_t\,)\;U(t)\,\phi(\mathbf{q},\,0\,)\,]\big|_x$$

that is

$$\frac{d}{ds}\,U(s)\big|_t = \frac{d}{dx}\,[\,\exp(\,\frac{x}{i}\,H_t\,)\;U(t)\,]\big|_0 \qquad \text{weakly on } \mathcal{D}\;, \tag{3.9}$$

or also $\frac{1}{i}\,H_t = \frac{d}{ds}\,U(s)\,U(t)^{-1}\,\big|_t$ weakly on $\mathcal{D}$, in formal correspondance to (3.3).

4. The elementary case $G = \mathbb{R}_+$

Let $T \in \mathbb{R}_+ \cup \{+\infty\}$ and $\mathcal{I}$ be the interval $[\,0, T\,[$. Also, let $\mathcal{C}$ be the convex cone of *strictly positive* functions in $C^\infty(\mathcal{I})$. Trivially, $\mathcal{C}$ is an abelian Lie group under function multiplication and the constant functions in $\mathcal{C}$ form an abelian subgroup $Const^*_{\mathcal{I}} \subset \mathcal{C}$. Therefore the factor group $\mathcal{C}_1 := \mathcal{C}/Const^*_{\mathcal{I}}$ is well-defined and is via the map $f \mapsto f/f(0)$ isomorphic to the convex subgroup of $\mathcal{C}$ in which all functions have initial value equal to one. We can thus identify $\mathcal{C}_1$ with $\{\, f \in \mathcal{C} \,|\, f(0) = 1 \,\}$ and denote by $\pi_1 : f \mapsto f/f(0)$ the pertinent projection. In the same manner, $C^\infty(\mathcal{I})/Const_{\mathcal{I}}$ is identified with the subgroup $C^\infty_0(\mathcal{I}) := \{f \in C^\infty(\mathcal{I}) \,|\, f(0) = 0\}$ of the additive group $C^\infty(\mathcal{I})$, the pertinent projection being $\pi_0 : f \mapsto f - f(0)$.

Let $f \in \mathcal{C}$ and $t \in \mathcal{I}$. By Theorem 1, the unique function $Ef_t(x) : \mathbb{R} \to \mathbb{R}_+$ satisfying (T1-1) is the exponential derivative of f in t, i.e. $Ef_t(x) = \exp\,[\,x\, f(t)^{-1} \frac{d}{ds} f(s)\big|_t\,] = \exp\,[\,x\, \frac{d}{ds} \log f(s)\big|_t\,]$ and by the Corollary to (T2-3), the mapping

$$\begin{aligned} E \,:\, \mathcal{C} &\longrightarrow \mathcal{C} \\ f &\longmapsto Ef_t := Ef_t(1) \end{aligned}$$

is an endomorphism of $\mathcal{C}$. In the following we continue to write simply Ef_t for $Ef_t(1)$ and agree upon these

Definitions

1. The unique operator $J : \mathcal{C} \longrightarrow \mathcal{C}_1$, $f \longmapsto Jf$, such that (EI)
$$E \circ J : \mathcal{C} \longrightarrow \mathcal{C} = Id_{\mathcal{C}} : f \longmapsto f \quad \text{(i.e. the identity of } \mathcal{C}\,),$$
is called the *exponential integral* **on** $\mathcal{C}$.
Notice this implies that E is *surjective* and J is *injective*.

2. Let $a, b \in \mathcal{I}$, $f \in \mathcal{C}$, and $J : \mathcal{C} \to \mathcal{C}_1$ as in (EI). (EIab)
The *exponential integral of* f *over the interval* $[\,a\,,b\,]$ is defined as
$$Jf\,|_a^b := Jf_b \,/\, Jf_a \qquad (\,=\, Jf(b)\,/\,Jf(a)\,).$$

3. Finally, under the same conditions as in (EIab) and additionally $a < b$, we call (EM)
$$\bar{f}_{[a,b]} := (\,Jf\,|_a^b\,)^{\frac{1}{b-a}}$$
the *exponential mean* of f over $[\,a\,,\,b\,]$.

The next theorem affirms that the just defined objects exist and that they are unique, i.e. they are well-defined.

THEOREM 3 (EXISTENCE AND UNIQUENESS OF Jf ETC.)

1. There exists a unique exponential integral $J : \mathcal{C} \longrightarrow \mathcal{C}_1$ as defined in (EI) and that is
$$Jf_t \;=\; \exp \int_0^t \log f(\tau)\, d\tau \tag{T3-1}$$

2. The exponential integral over $[\,a\,,\,b\,] \subset \mathcal{I}$ equals
$$Jf\,|_a^b = \exp \int_a^b \log f(\tau)\, d\tau\,. \tag{T3-2}$$

3. Also, the exponential mean of f over $[\,a\,,\,b\,]$ is
$$\bar{f}_{[a,b]} \;=\; \exp \frac{1}{b-a} \int_a^b \log f(\tau)\, d\tau\,. \tag{T3-3}$$

Proof: Trivially, the composition of

$$\begin{aligned} E \,:\, \mathcal{C} &\longrightarrow \mathcal{C} \\ f(t) &\longmapsto Ef_t = \exp \frac{d}{dt} \log f(t) \end{aligned} \qquad \text{with} \qquad \begin{aligned} J \,:\, \mathcal{C} &\longrightarrow \mathcal{C}_1 \\ f(t) &\longmapsto Jf_t = \exp \int_0^t \log f(\tau) d\tau \end{aligned}$$

assigns to every $f \in \mathcal{C}$

$$E \circ J\,(f) \;=\; \exp \frac{d}{dt} \log \exp \int_0^t \log f(t) d\tau \;=\; f\,.$$

Hence $E \circ J \equiv Id_{\mathcal{C}}$. Similarly it is seen that $J \circ E \restriction \mathcal{C}_1 \equiv Id_{\mathcal{C}_1}$. (4.1)

Furthermore J is an exponential integral as defined in (EI) and J is unique. In fact, by (T2-2) $E : f \mapsto Ef$ is homomorhic, i.e. $E(fg) = (Ef)\cdot(Eg)$, for every $f, g \in \mathcal{C}$. Moreover, $E \restriction \mathcal{C}_1$ is *injective* due to (4.1), as is also $J : \mathcal{C} \to \mathcal{C}_1$. It follows that for every $f \in \mathcal{C}_1$, $Ef \equiv 1$ if and only if $f \equiv 1$, and that for any J_1 and J_2 as in definition (EI) and $g \in \mathcal{C}$

$$\begin{aligned} E \circ J_1 g = E \circ J_2 g &\iff E \circ J_1 g \,/\, E \circ J_2 g \equiv 1 \\ &\iff (E \restriction \mathcal{C}_1)(J_1 g / J_2 g) \equiv 1 \\ &\iff J_1 g \,/\, J_2 g \equiv 1, \\ &\iff J_1 g = J_2 g, \end{aligned}$$

which can only apply to every $g \in \mathcal{C}$ if $J_1 \equiv J_2$.
The remainder is just as straightforward.

Note that identities (4.1) imply that $E \restriction \mathcal{C}_1 : \mathcal{C}_1 \to \mathcal{C}$ and $J : \mathcal{C} \to \mathcal{C}_1$ are *bijective* mappings, i.e. they are *isomorphisms*. Obviously this is not the case for $E : \mathcal{C} \to \mathcal{C}$, since $Ef \equiv E(\alpha f)$, for every $\alpha \in \mathbb{R}_+$, in particular $Ef \equiv E(f/f(0))$.

The formal correspondance between the classical and the exponential derivative is already seen in some simple examples:

$D = \frac{d}{dt} : C^\infty(\mathcal{I}) \to C^\infty(\mathcal{I})$	$E : \mathcal{C} \to \mathcal{C}$
$D(const) \equiv 0$	$E(const) \equiv 1$
$D(\alpha f + \beta g) = \alpha Df + \beta Dg$	$E(f^\alpha \cdot g^\beta) = (Ef)^\alpha \cdot (Eg)^\beta$
in particular: $D(\alpha + f) = Df$	$E(\alpha \cdot f) = Ef \quad (\alpha \in \mathbb{R}_+)$

Further properties are listed in the next

THEOREM 4 (DERIVATION RULES)
Let $f, g \in \mathcal{C}$ and as above D denote $\frac{d}{dt}$.
These rules are easily verified:

1. $E(f+g) = (Ef)^{\frac{f}{f+g}} \cdot (Eg)^{\frac{g}{f+g}}$,
2. $E(f^g) = (Ef)^g \cdot f^{Dg}$,
3. $E(f \circ g) = [(Ef) \circ g]^{Dg}$ if $g(\mathcal{I}) \subset \mathcal{I}$,
4. $E(E(f)) = [\,E(\frac{Df}{f})\,]^{\frac{Df}{f}}$ if $Df \in \mathcal{C}$,
5. $\frac{f(b)}{f(a)} = (\bar{E}f_{[a,b]})^{(b-a)}$, for $[\,a\,,\,b\,] \subset [\,0\,,\,T\,[$.

Proof: Mere retrospection of the definitions and Theorem 3. We show the 1st and 5th statements and leave the remainder to the reader.

Ad 1. :

$$\begin{aligned} E(f+g) &= \exp D \log(f+g) = \exp \frac{Df+Dg}{f+g} \\ &= \exp \frac{Df}{f+g} \cdot \exp \frac{Dg}{f+g} = (\exp \frac{Df}{f})^{\frac{f}{f+g}} \cdot (\exp \frac{Dg}{g})^{\frac{g}{f+g}} \\ &= (Ef)^{\frac{f}{f+g}} \cdot (Eg)^{\frac{g}{f+g}} \end{aligned}$$

Ad 5. :

$$\frac{f(b)}{f(a)} = \frac{\pi_1 f(b)}{\pi_1 f(a)} \underset{(\wr\,(4.1))}{=} \frac{J\,E\pi_1 f(b)}{J\,E\pi_1 f(a)} \underset{(\wr\,E\pi_1 f = Ef)}{=} \frac{J\,Ef(b)}{J\,Ef(a)} \underset{(\wr\,\mathrm{EIab})}{=} J\,Ef\,|_a^b \underset{(\wr\,\mathrm{EM})}{=} (\bar{E}f_{[a,b]})^{(b-a)}$$

In the Appendix, these concepts are applied to earthly reality in a mathematical model of epidemic spread.

5. Appendix

A 'real world' problem

In the model of *epidemic spread* presented here, we consider strictly positive C^∞-functions on a time interval $\mathcal{I} := [\,0\,,\,T\,[\;,\; T \in \mathbb{R}_+ \cup \{+\infty\}$. Remember these functions form a convex cone and a multiplicative group $\mathcal{C} \subset C^\infty(\mathcal{I})$ and $\mathcal{C}_1 = \{\, f \in \mathcal{C} \,|\, f(0) = 1 \,\}$ is a convex subgroup of $\mathcal{C}$. Also, it is easily seen that

$$E \circ J \equiv Id_{\mathcal{C}} \quad \textit{and} \quad J \circ E \restriction \mathcal{C}_1 \equiv Id_{\mathcal{C}_1} \qquad [\text{ cf. (4.1) }]\,, \tag{A.1}$$

where $E : \mathcal{C} \to \mathcal{C}$ and $J : \mathcal{C} \to \mathcal{C}_1$ respectively are the mappings

$$Ef_t = \exp \frac{d}{ds} \log f(s)\,\big|_t \quad \text{and} \quad Jf_t = \exp \int_0^t \log f(s)\, ds\,. \tag{A.2}$$

Remember also that the exponential integral and the exponential mean of f over $[\,a\,,\,b\,] \subset \mathcal{I}$ respectively equal

$$Jf\,|_a^b = \exp \int_a^b \log f(s)\, ds \quad \text{and} \quad \bar{f}_{[a,b]} = \exp \frac{1}{b-a} \int_a^b \log f(s)\, ds\,. \tag{A.3}$$

For simplicity, the *herd size* is normalized to one in the following. The non-susceptible ('immune') fraction of the herd is represented by a function $h : \mathcal{I} \to [\,0\,,\,1\,]$ (0: no immune members, 1: total herd immunity). More precisely, $h(t)$ is defined as the *unsusceptible* fraction, comprising the immune members, but also those who are currently sick, and those who have died, i.e. $h(t) = 1 - s(t)$, $s(t)$ denoting the susceptible fraction. (Note that in our model, $h(t) = 1 - s(t)$ is *not equal* to the sum $i(t)+r(t)$ of the infected and recovered fractions, as is the case for example in the SIR model, cf. [SM] , since we do not require that recovered members remain immune.)

Finally, the *incidence*, defined as the fraction of *symptomatic* ('acute') new infections per time, is represented by the function $u : \mathcal{I} \to \mathbb{R}_+$.

The *reproduction factor* indicates how many members are infected on average by one contagious member. By itself, it says thus nothing about the underlying time scale. Yet time enters naturally with the *transmission time* T_{tr} which is the average time during that one sick member infects R susceptible ones. Then obviously $u(\,t + T_{tr}\,) = R \cdot u(\,t\,)$ and we can state the following

Proposition 1 (Reproduction factors)
With Eu_t as above, the reproduction factor, usually defined as $R := \frac{u(\,t+T_{tr}\,)}{u(\,t\,)}$, equals:

$$R = (\,\bar{Eu}_{[t,t+T_{tr}]}\,)^{T_{tr}}, \tag{A.4}$$

i.e. $R^{\frac{1}{T_{tr}}}$ is the *exponential mean* of Eu over the transmission interval $[\,t,\, t + T_{tr}\,]$.

Proof : Immediate consequence of Theorem 3 (T3-3).

Remarks

1. Since $Eu : t \mapsto Eu_t$ is continuous, there always exists a time $\tau \in [\,t\,,\,t + T_{tr}\,]$, such that $Eu_\tau = \bar{Eu}_{[t,t+T_{tr}]}$, and hence, due to (A.4),

$$R^{\frac{1}{T_{tr}}} = Eu_\tau \tag{A.5}$$

2. Trivially, the transmission time T_{tr} introduces a *natural time scale* in which T_{tr} is the unit of time. Hence in this scale, for a certain $\tau \in [\,t\,,\,t+1\,]$,

$$R = Eu_\tau\ . \tag{A.6}$$

From now on we work in this natural time scale, i.e. with $T_{tr} = 1$. Moreover, we identify $R = R_t$ with $Eu_t = \exp \frac{d}{dt} \log u(t)$, which amounts to refining the definition of R by replacing the exponential mean of Eu over a transmission cycle by its *current* value at time t. This exactly is summarized in the following

Proposition 2 . In the natural time scale (in which $T_{tr} = 1$) , the reproduction factor is the exponential derivative of the incidence.

Every mathematical model makes simplifications. An essential assumption of ours is a *homogeneous environment*. This means, to put it simply, that the pathogene finds the same conditions for spreading throughout the herd. In particular, singular spreading events or new infections from outside are neglected. As far as these conditions are met, the essence of the disease spread is captured by the following *model equations*, which relate the immune and fatal fractions to the incidence

$$h(t) \;=\; h(0) \;+\; w_i \int_0^t \exp\,(\frac{\tau - t}{T_i})\, u(\tau)\, d\tau \;+\; w_f \int_0^t u(\tau)\, d\tau \;, \qquad \text{(A.7)}$$
$$\text{with}\, w_i \;=\; \frac{B}{100 - A} \quad \text{and}\, w_f \;=\; \frac{C}{100 - A}\,.$$

The first integral represents the recovered uninfectable ('steril') fraction and the second integral stands for the deceased part of the herd at time t. The time constant T_i limits the (mean) immunity duration after infection, and the weighting factors w_i and w_f introduce the percentage A of the *asymptomatic* ('silent') cases, B of *steril-immunized* cases, and C of *fatal* cases into the equation (so if all infected survivors become non-infectiously immune, then obviously $B + C = 100$).
Under the assumed homogeneous conditions, the exponential derivative of the incidence is directly proportional to the susceptibility. By virtue of proposition 2, this means

$$\begin{aligned} R_t \;&=\; Eu_t \;=\; s(t)\, Eu_0 \;=\; R_0\,[\,1 \;-\; h(t)\,] \\ &=\; R_0\,[\,1 - h(0) - w_i \int_0^t \exp\,(\frac{\tau - t}{T_i})\, u(\tau)\, d\tau \;-\; w_f \int_0^t u(\tau)\, d\tau\,]\,, \end{aligned} \qquad \text{(A.8)}$$

and so the incidence becomes, in virtue of (A.1), (EI), and (A.8)

$$\begin{aligned} u(t) \;&=\; u(0)\; J\, Eu_t \;=\; u(0)\, \exp \int_0^t \log Eu_\tau\, d\tau \\ &\overset{\text{cf. (A.1)}}{=} \; u(0)\; \exp \int_0^t \log\{\, R_0\,[\,1 - h(0) - w_i \int_0^\tau \exp\,(\frac{\theta - \tau}{T_i})\, u(\theta)\, d\theta \;-\; w_f \int_0^\tau u(\theta)\, d\theta\,]\,\}\; d\tau\,, \end{aligned} \qquad \text{(A.9)}$$

where $u(0)$, $h(0)$ and R_0 represent respectively the initial incidence, the initial herd infection and the reproduction factor at time $t = 0$.

Equation (A.9) describes the development of an epidemic under homogeneos conditions with no temporal changes others than the epidemic spread. The exorbitant dynamics that an epidemic can develop at higher reproduction factors is also seen in the derivative of $u(t)$ obtained from (A.9)

$$\frac{du}{dt}\Big|_t \;=\; u(t)\; \log\{\, R_0\,[\,1 - h(0) - \; w_i \int_0^t \exp\,(\frac{\tau - t}{T_i})\, u(\tau)\, d\tau \;-\; w_f \int_0^t u(\tau)\, d\tau\,]\,\}\,. \qquad \text{(A.10)}$$

We have integrated equation (A.9) for basic reproduction factors varied from $R_0 = 0.5$ to $R_0 = 3.0$ using a numerical routine implemented in the AMADEUS program, cf. [Hei] . The initial conditions are $h(0) = 0$ and $u(0) = 10^{-5}$ (in common parlance: 'incidence one', i.e. one sick person in a population of 100.000), and we assume $T_i = \infty$ and $B + C = 100$ percent, that is every surviving infected member ends up permanently immune. We also allow $A = 20$ percent asymptomatic cases.

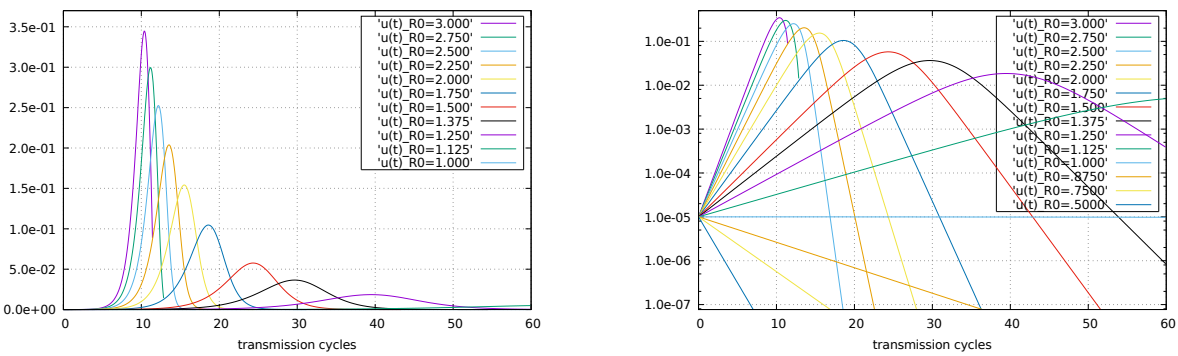

Fig. 1. *The incidence*

The graphics give an idea of what is happening if an epidemic with higher reproduction factor is let loose, i.e. if it is not contained by suitable measures. For $R_0 \gtrsim 2.7$, a considerable part of the population would get simultaneously sick at the peak.

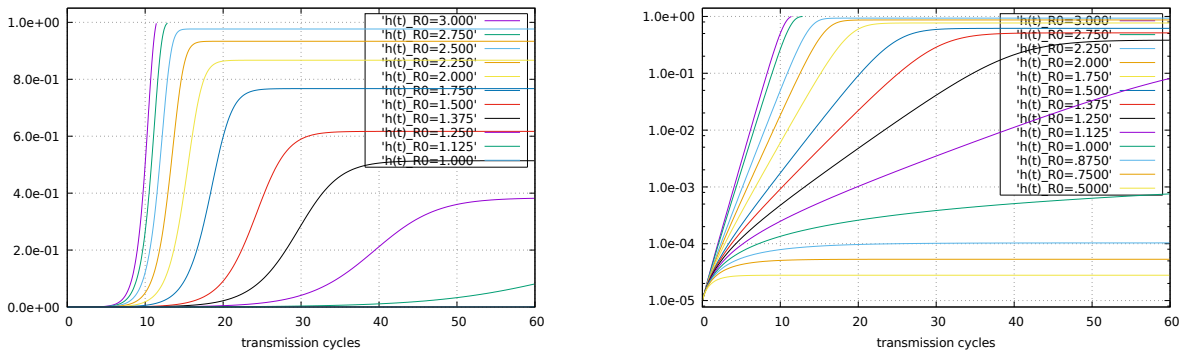

Fig. 2. *The herd infection*

With initially $R_0 \gtrsim 2.7$ and without further mesures, everybody would eventually go through the infection, be it alive or dead. The entire population would end up immune (the cuts in the two upper curves indicate when 100 % herd immunity is attained). Assuming a mortality rate of 0.2 %, for example, in a population of 82 million, as in Germany, more than 160,000 people would die. Under such conditions one can only hope for any effective vaccination, otherwise the reproduction factor would have to be kept low with painful containment measures.

6. Conclusions and outlook

The exponential derivative of a differentiable path $c : \mathcal{I} \to G$ in a Lie group has been studied and it was demonstrated how it acts *within* the group through right or left multiplication and in this way provides an *intrinsic* description of the infinitesimal change, in contrast to the classical differential that acts in *tangent* space.

Basic properties have been outlined and calculation rules derived. The elementary case of the multiplicative group $G = \mathbb{R}_+$ together with the smooth strictly positive functions $f : \mathcal{I} \to \mathbb{R}_+$ was treated in more detail, and applied to a numerical model of epidemic spread.

We are particularly interested in novel methods and applications in dynamic systems with many interacting components. The general framework in finite or infinite dimensional Lie groups may be of interest here. However, this must be reserved for further investigations.

References

[Ddn] Dieudonné, J., *Treatise on Analysis*, Chap 16: *Differentiable Manifolds*, Academic Press, 1972

[Jst] Jost, J., *Riemannian Geometry and Geometric Analysis*, Second Edition, Springer-Verlag, 1998

[Hel] Helgason, S. *Differential Geometry, Lie Groups, and Symmetric Spaces*, Academic Press, 1978

[Bai] Bailey, N. T. J., *The Mathematical Theory of Infectious Diseases* (2nd Edition), Hafner, New York, 1975

[SM] Smith, D., Moore, L., *The SIR Model for Spread of Disease - The Differential Equation Model*, MAA Publications, Convergence, December 2001, `https://www.maa.org/press/periodicals/loci/joma/the-sir-model-for-spread-of-disease-the-differential-equation-model`

[vDr] Van den Driessche, P., *Reproduction numbers of infectious disease models*, Infectious Disease Modelling, Volume 3, issue 4, August 2017, pp. 288-302

[Hei] Hein, S., AMADEUS: *A plain numerical Model Approximating the Development of Epidemics Under varied conditions of Spread*, `https://github.com/SteffenHein/amadeus.git`

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